

Detecting Synthetic Music using a Hybrid Transformer–Autoencoder Framework

Generative AI Project Proposal

Group 28

Minahil Ali (22I-0849)

Ayaan Khan (22I-0832)

Atif Ibrahim Abasi (22I-1249)

October 19, 2025

1 Introduction & Motivation

The surge of AI-generated music powered by models like MusicLM [1], Riffusion[5], and Mubert[9] has blurred the boundaries between human and machine creativity. As generative audio systems become more accessible, they pose new challenges in authenticity verification, copyright enforcement, and digital media integrity. While synthetic image and text detection have seen extensive study[15, 12, 14], the detection of AI-composed music remains largely unexplored.

This project aims to address that gap by developing a hybrid generative–transformer-based framework to detect AI-generated music. Unlike traditional CNN classifiers, which often rely on shallow spectral features, our proposed model integrates Transformer-based semantic embeddings with Autoencoder-based generative reconstruction signatures to identify subtle differences between human and AI music compositions.

The goal is not only to build a functioning detector but to analyze how generative and attention-based architectures perceive musical authenticity, contributing novel insights to the field of Generative Audio Forensics.

2 Problem Statement

Modern AI music models generate audio that mimics the structure, tone, and rhythm of real compositions[3, 10, 4] , making manual or rule-based detection infeasible[2].

Research Question: Can we combine transformer-based embeddings and generative autoencoder features to robustly detect synthetic music created by different AI models?

We hypothesize that transformer encoders can capture high-level musical semantics (style, emotion, rhythm), while autoencoders capture low-level generative patterns. Their fusion offers complementary insights, enabling a robust and generalizable detector.

3 Objectives

- Develop a hybrid model combining Transformer-based embeddings and Autoencoder latent features to classify AI vs. human-composed music.
- Compare this hybrid approach against individual architectures: CNN-only, Transformer-only, and Autoencoder-only.
- Evaluate the system across multiple AI-generated music datasets (MusicLM, Mubert, Riffusion).
- Conduct a comparative study and visual analysis of latent representations and reconstruction patterns.
- Demonstrate the generalization of the detector on unseen generative music sources.

4 Proposed Methodology

4.1 Dataset and Preprocessing

The dataset will consist of AI-generated samples (MusicLM, Mubert, Riffusion) and human-composed samples (MAESTRO[7], EMOPIA[8], Lakh MIDI[13]). All samples will be converted into Mel-spectrograms (128 mel bands, 5s duration, 16kHz). Data augmentation (pitch shifting, time stretching) ensures balance across classes.

4.2 Model Architecture

1. **Transformer Embeddings:** Use a pretrained CLAP (Contrastive Language-Audio Pretraining)[4] or Audio Spectrogram Transformer[6] (AST) model to extract 512-

dimensional semantic embeddings from Mel-spectrograms. These embeddings capture stylistic and emotional cues from the music.

2. **Generative Autoencoder:** Train a Convolutional Autoencoder (CAE)[11] on human-composed music to learn the underlying data distribution. Extract the latent vector (256 dimensions) and the reconstruction loss as indicators of “realness.”
3. **Feature Fusion and Classification:** Concatenate Transformer embeddings and Autoencoder latent vectors into a unified feature representation. Pass this combined feature through a Multi-Layer Perceptron (MLP) classifier to predict whether a sample is AI-generated or human-made.
4. **Comparative Baselines:** Implement CNN-only classifier (simple baseline), Transformer-only (fine-tuned CLAP/AST), and Autoencoder-only anomaly detector variants.
5. **Evaluation:** Evaluate all models using Accuracy, Precision, Recall, F1-score, ROC-AUC, and cross-model generalization (e.g., trained on MusicLM, tested on Riffusion). Visualize feature embeddings via t-SNE and reconstruction error heatmaps to interpret model behavior.

5 Tools and Frameworks

- **Programming:** Python
- **Deep Learning:** PyTorch, Hugging Face Transformers, torchaudio
- **Pretrained Models:** CLAP, AST
- **Data Handling:** Librosa, pandas, numpy
- **Visualization:** Matplotlib, Seaborn, TensorBoard
- **Hardware:** GPU-enabled Colab or Kaggle environment

6 Expected Outcomes

- A hybrid Transformer–Autoencoder detector capable of distinguishing human vs. AI-generated music.
- Comparative results showing superior generalization of the hybrid model over baseline architectures.

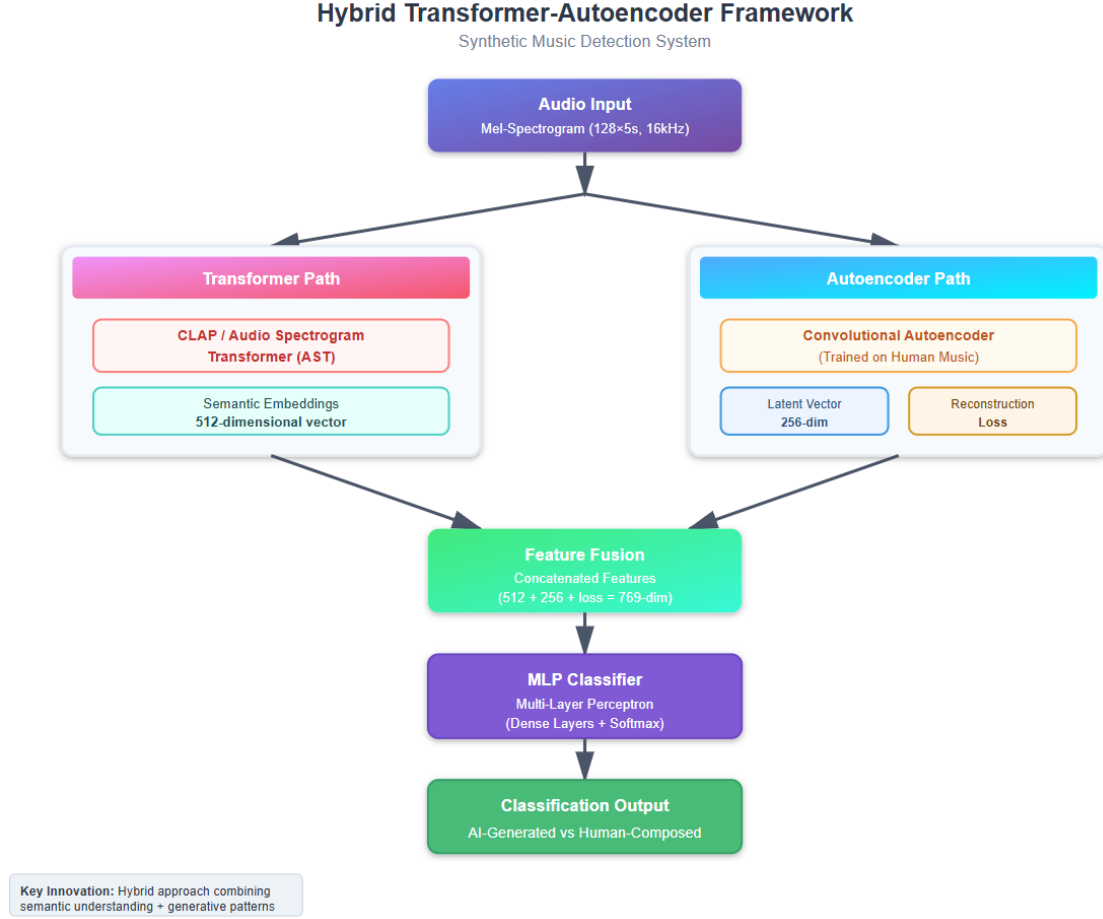


Figure 1: Proposed Hybrid Transformer–Autoencoder Framework

- Visual and statistical evidence of distinct latent patterns between synthetic and human music.
- A research report contributing to AI music forensics and authenticity detection.

7 Novelty and Research Contribution

- **Novelty:** Very few studies target synthetic music detection[14] ; introduces a hybrid generative-transformer approach.
- **Generative AI Alignment:** Integrates Autoencoder and Transformer—core Generative AI architectures.
- **Comparative Analysis:** Evaluates multiple model families (CNN, AE, Transformer, Hybrid) on multiple datasets.

- **Research Impact:** Opens a new direction in audio authenticity detection, complementing deepfake and image forensics.

Related Work

Existing studies in generative audio have primarily focused on music creation and representation learning [2, 3, 10, 4]. In contrast, research on detecting synthetic audio remains limited [12, 14]. Our work builds upon these foundations by integrating transformer-based semantic features and autoencoder generative signatures for AI music authenticity detection.

References

- [1] Andrea Agostinelli, Timo I Denk, et al. Musiclm: Generating music from text. *arXiv preprint arXiv:2301.11325*, 2023.
- [2] Jean-Pierre Briot, Gaëtan Hadjeres, and François-David Pachet. *Deep Learning Techniques for Music Generation*. Springer, 2020.
- [3] Prafulla Dhariwal, Heewoo Jun, Christine Payne, and et al. Jukebox: A generative model for music. *OpenAI Blog*, 2020.
- [4] Benjamin Elizalde, Cheng Wang, Goel Kratarth, Eduardo Fonseca, et al. Clap: Learning audio concepts from natural language supervision. *IEEE Transactions on Audio, Speech, and Language Processing*, 2023.
- [5] Seth Forsgren and Hayk Martiros. Riffusion: Stable diffusion for real-time music generation. <https://www.riffusion.com/about>, 2022.
- [6] Yuan Gong, Yu-An Chung, and James Glass. Ast: Audio spectrogram transformer. In *Interspeech*, pages 571–575, 2021.
- [7] Curtis Hawthorne, Erich Elsen, et al. Enabling factorized piano music modeling and generation with the maestro dataset. In *International Conference on Learning Representations (ICLR)*, 2019.
- [8] Ching-Hua Hung, Szu-Yu Chou, and Yi-Hsuan Chen. Emopia: A multi-modal pop piano dataset for emotion recognition and emotion-based music generation. In *ISMIR*, 2021.
- [9] Mubert Inc. Mubert: Ai music generation platform. <https://mubert.com>, 2024.

- [10] Felix Kreuk and et al. Audiogen: Textually guided audio generation. *arXiv preprint arXiv:2301.12661*, 2023.
- [11] Alireza Makhzani, Jonathon Shlens, Navdeep Jaitly, Ian Goodfellow, and Brendan Frey. Adversarial autoencoders. *arXiv preprint arXiv:1511.05644*, 2015.
- [12] Parul Mittal and Pankaj Kumar. Survey on deepfake audio detection: Techniques and challenges. *IEEE Access*, 11:67842–67858, 2023.
- [13] Colin Raffel. Learning-based methods for comparing sequences, with applications to audio-to-midi alignment and matching. In *ISMIR*, 2016.
- [14] Jing Wang, Xin Li, and Zhi Chen. Towards synthetic audio detection: challenges in distinguishing ai-generated music. In *IEEE ICASSP*, 2023.
- [15] Yichen Zhang, Xuefeng Liu, and Jian Wang. A comprehensive survey of deepfake image detection. *ACM Computing Surveys*, 2023.